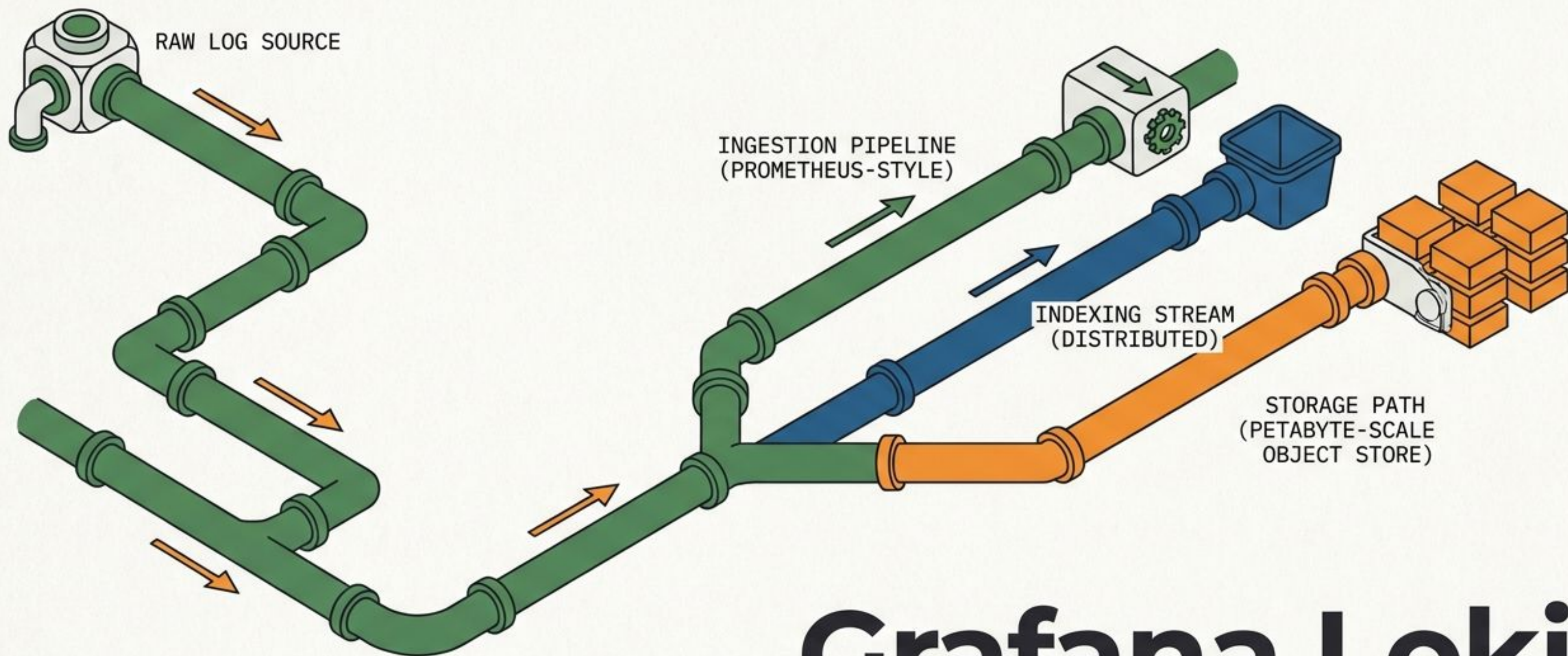
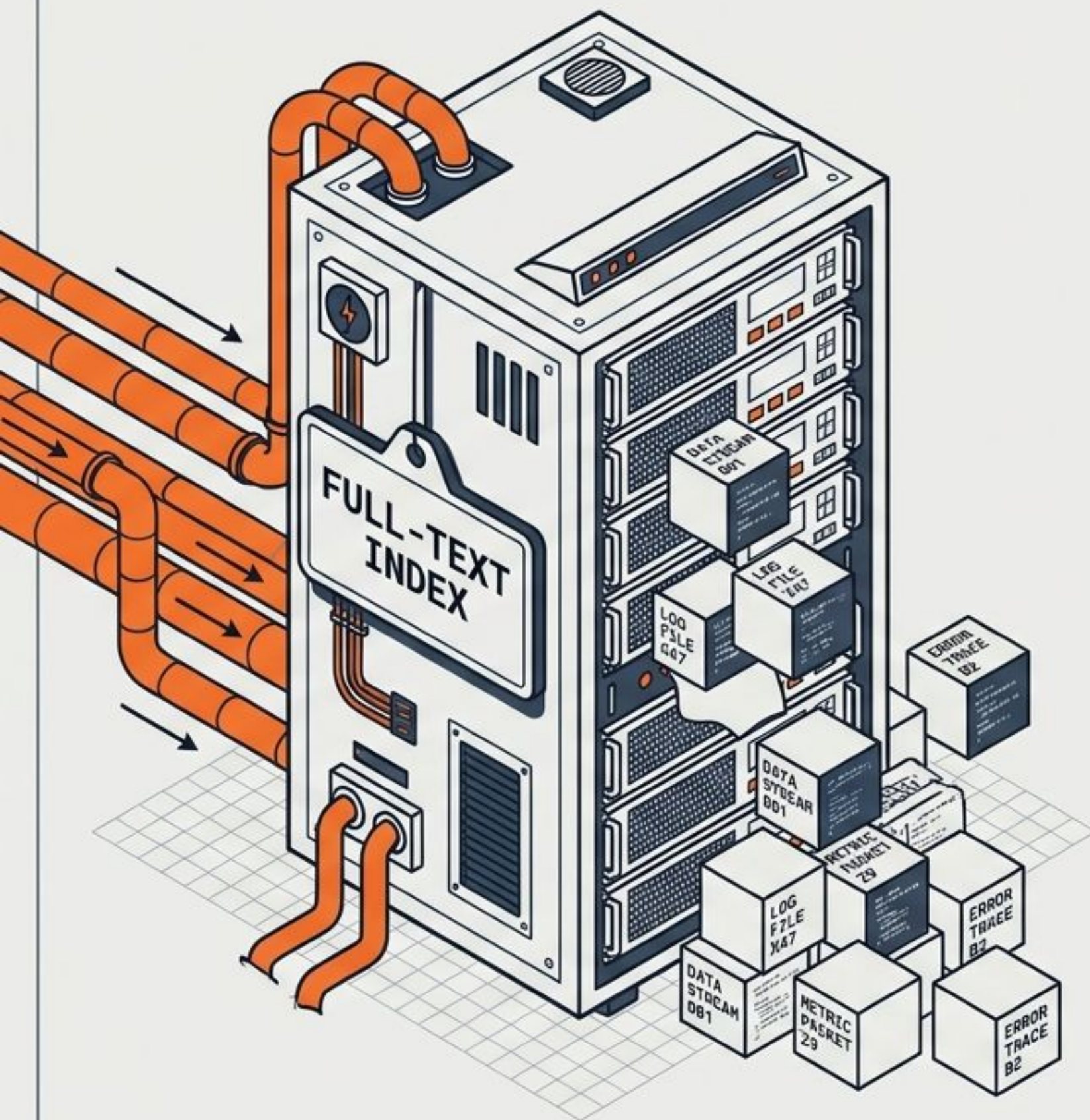




Grafana Loki

Architecting Petabyte-Scale
Log Aggregation



The Petabyte Wall

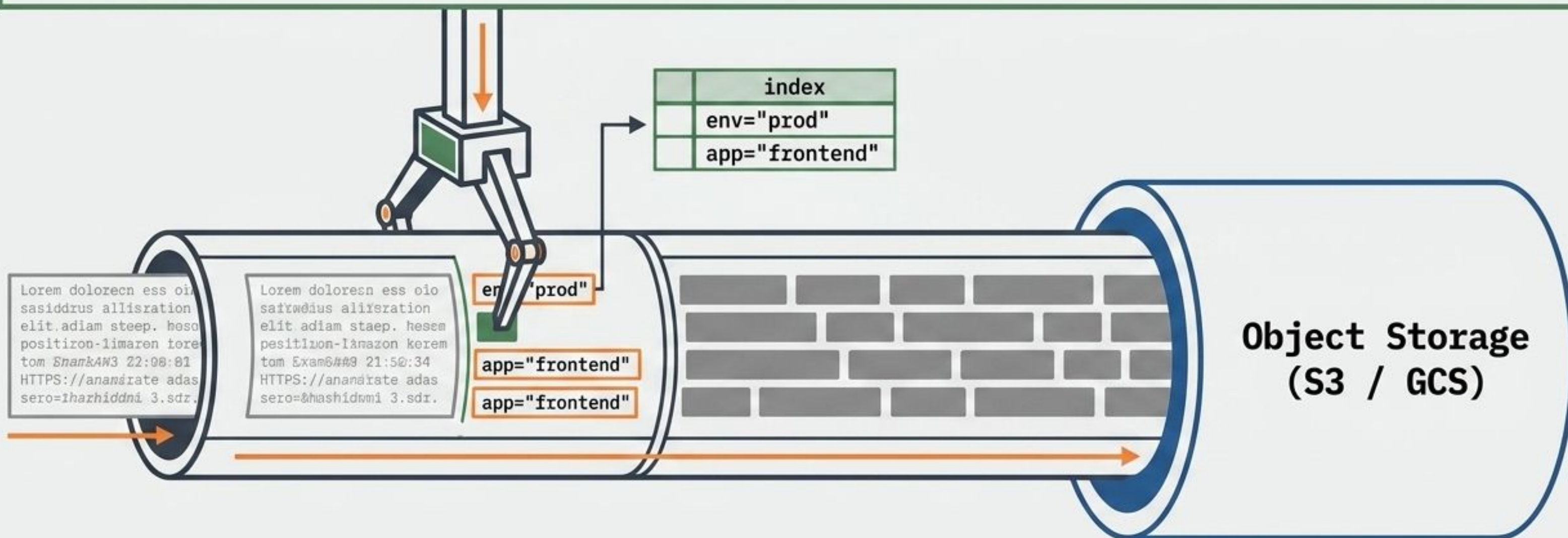
Traditional logging indexes every word of every log line. At petabyte scale, the index becomes larger than the data itself.



Scaling centralized, full-text logging is financially and computationally unsustainable.

Labels, Not Full-Text Indexes

**Loki does not index the contents of the logs.
It only indexes metadata (labels).**



The Anatomy of a Stream

1. Stream ID {tenant="A", app="payment", env="prod"}



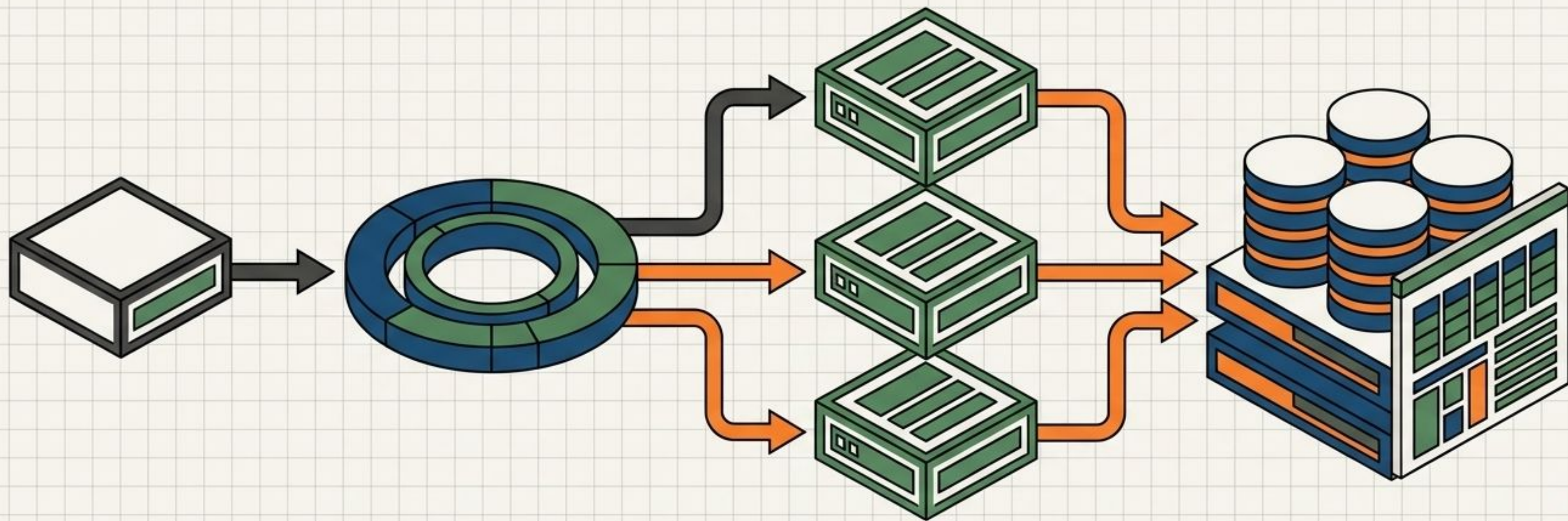
2. TSDB Index Maps exact label sets to the physical locations of stored data over time.



3. Chunk

A highly compressed container of raw log entries, nanosecond timestamps, and structured metadata for a specific stream within a distinct time window.

The Write Path: Fan-Out and Flush



Distributor

Validates and hashes the incoming log stream.

Hash Ring

Uses consistent hashing to find correct node owners.

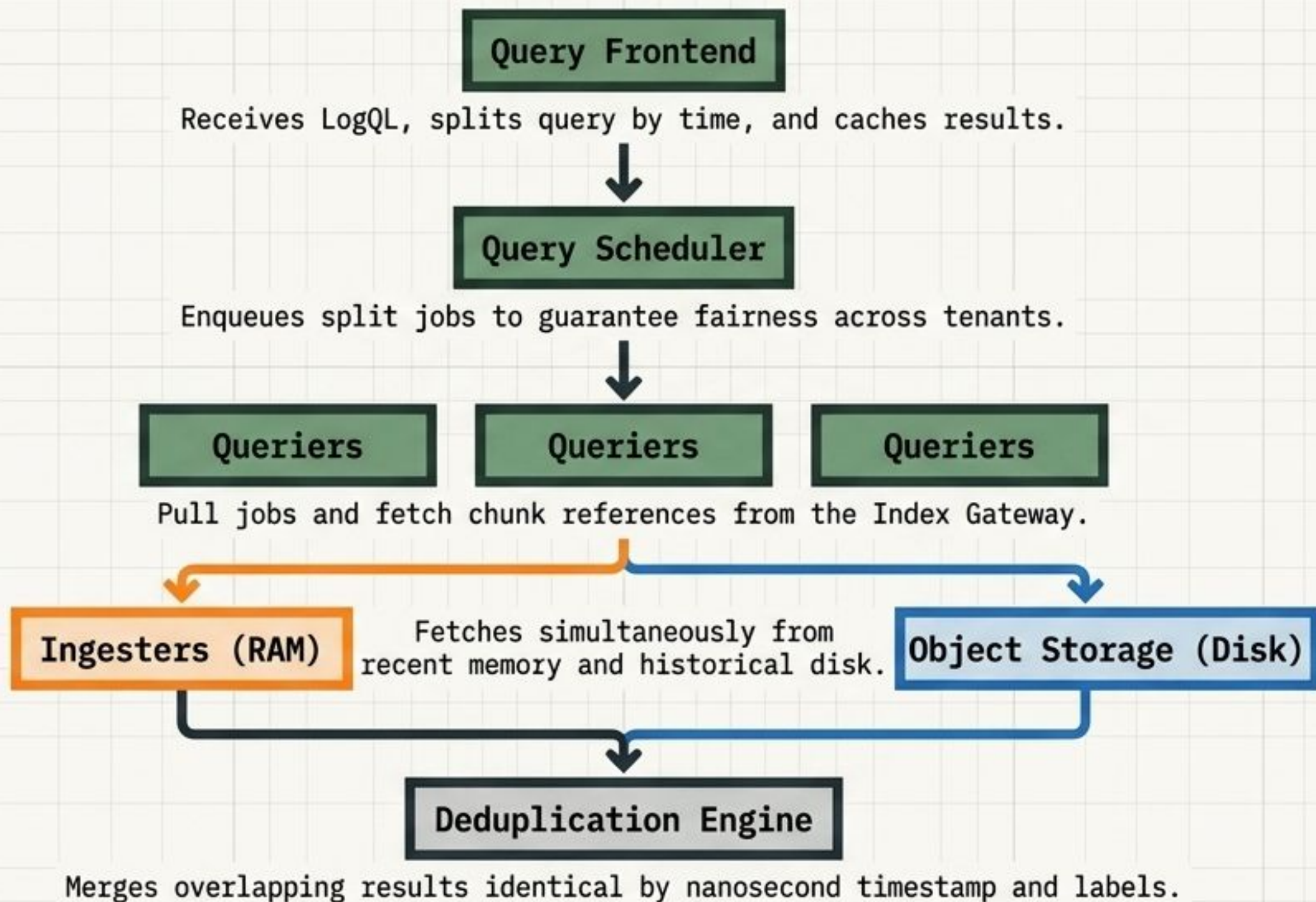
Ingesters

Replicates data to a quorum to prevent loss. Buffers log data heavily in active memory.

Object Storage & TSDB

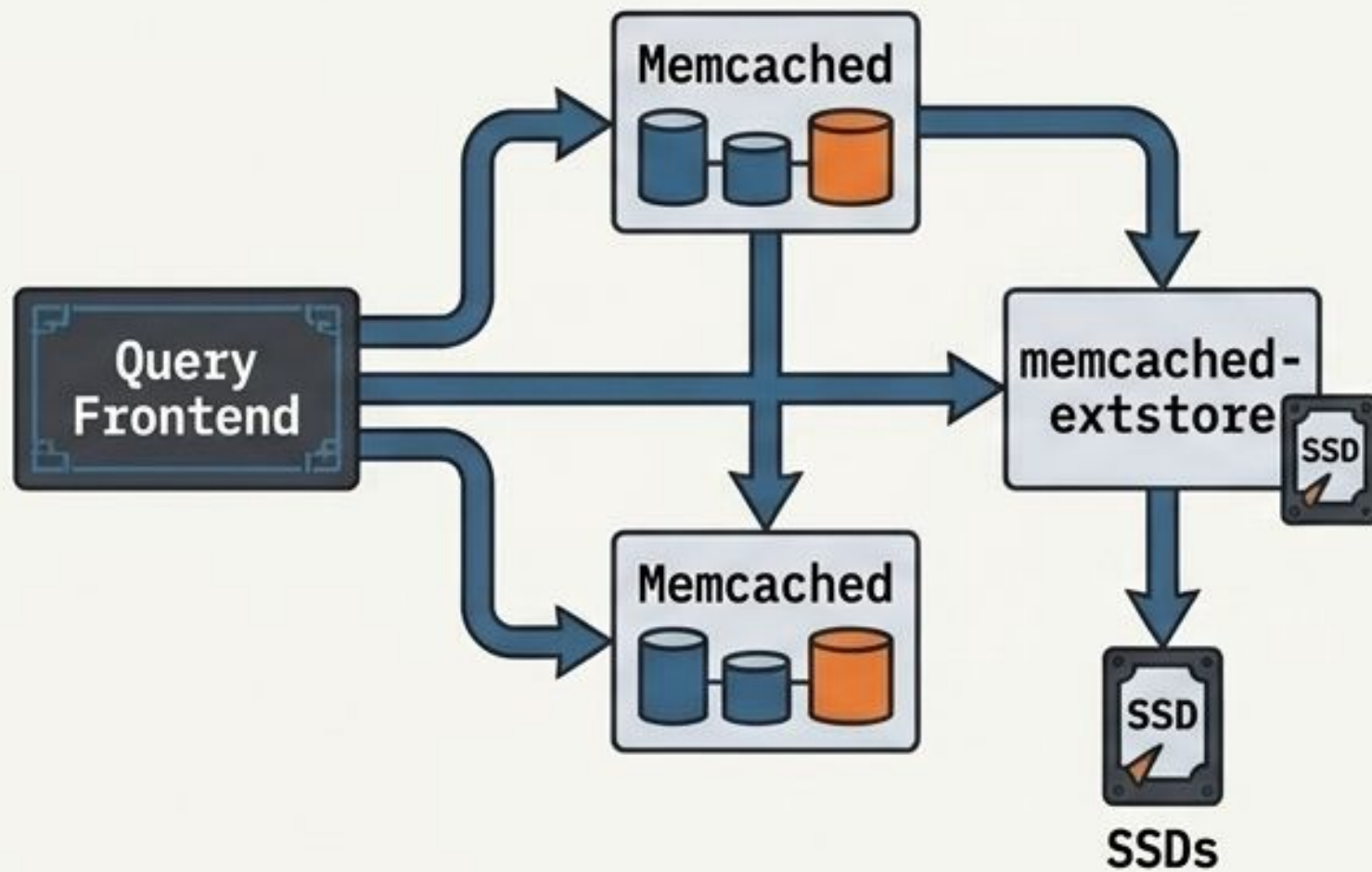
Flushes immutable chunks to disk and asynchronously updates the TSDB index.

The Read Path: Scatter and Gather



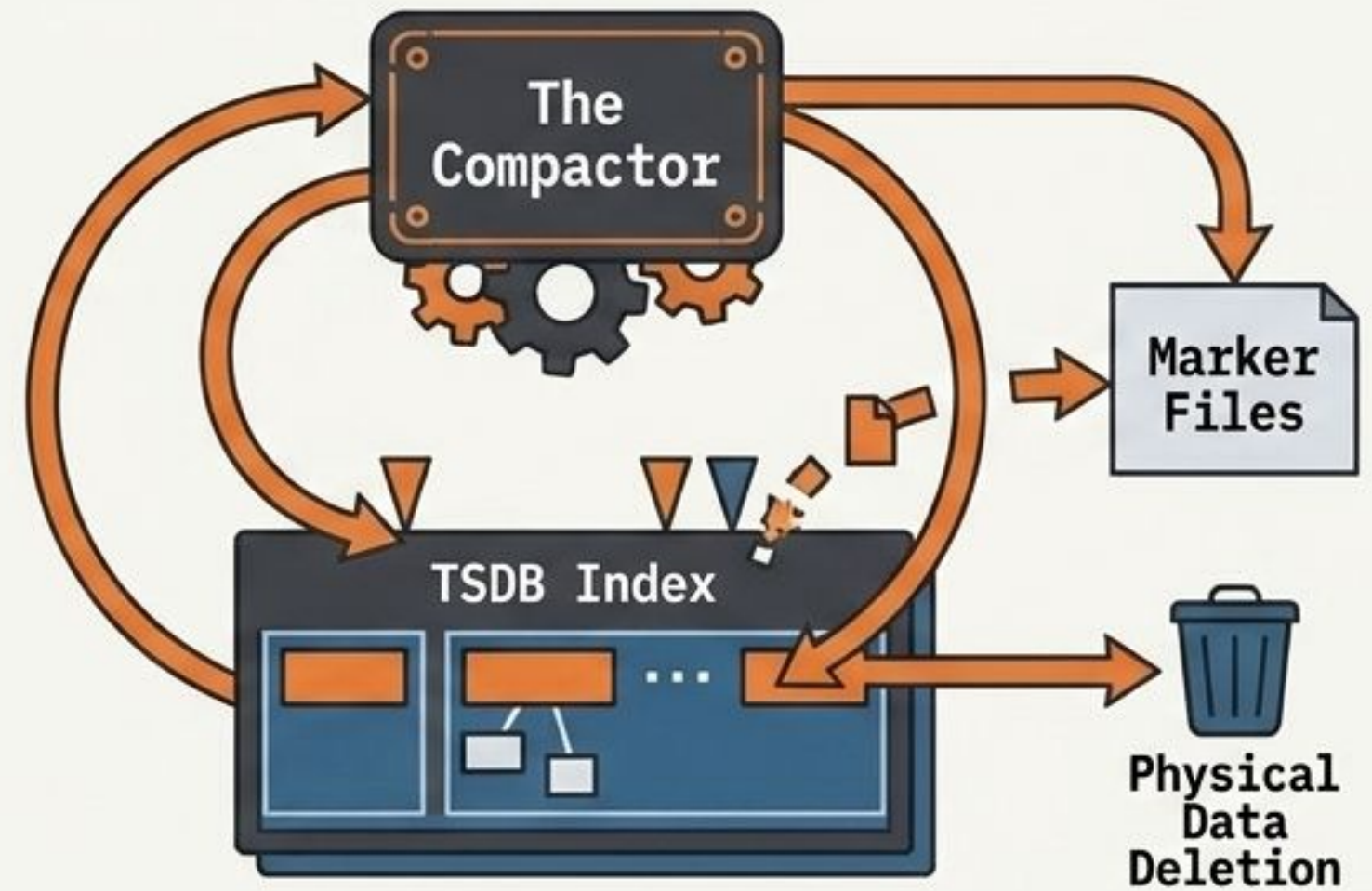
Optimization & Lifecycle Operations

Caching Sub-System



Multi-tiered caching (Results Cache & Chunks Cache). Uses 'memcached-extstore' to seamlessly scale cache capacity to attached SSDs for massive deployments.

Retention Sub-System



Retention is applied globally or per-tenant. The Compactor removes chunk references from the index and creates marker files; an asynchronous sweeper safely deletes the physical data later.